CSE400 – Fundamentals of Probability in Computing

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

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1 Marginal PMF

1.1 Definition

For discrete random variables X and Y, the joint PMF is:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The marginal PMFs are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$
$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

1.2 Marginal PMF from Joint PMF Table

- Row sums give $p_X(x)$
- Column sums give $p_Y(y)$

Mathematically:

$$p_X(x_i) = \sum_j p_{ij}$$
$$p_Y(y_j) = \sum_i p_{ij}$$

Normalization:

$$\sum_i p_X(x_i) = 1$$
$$\sum_j p_Y(y_j) = 1$$

1.3 Example

Given joint PMF:

	$y = 0$	$y = 1$
$x = 0$	$\frac{1}{8}$	$\frac{1}{4}$
$x = 1$	$\frac{3}{8}$	$\frac{1}{4}$

Step-by-step Solution with Logic

Step 1: Identify rows and columns

- Rows correspond to fixed x values
- Columns correspond to fixed y values

Step 2: Compute $p_X(x)$ (row-wise sum)

$$\begin{aligned} p_X(0) &= \frac{1}{8} + \frac{1}{4} \\ &= \frac{1}{8} + \frac{2}{8} \\ &= \frac{3}{8} \end{aligned}$$

$$\begin{aligned} p_X(1) &= \frac{3}{8} + \frac{1}{4} \\ &= \frac{3}{8} + \frac{2}{8} \\ &= \frac{5}{8} \end{aligned}$$

Logic: We sum across columns because we eliminate Y to get marginal of X.

Step 3: Compute $p_Y(y)$ (column-wise sum)

$$\begin{aligned} p_Y(0) &= \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2} \\ p_Y(1) &= \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \end{aligned}$$

Logic: We sum across rows because we eliminate X to get marginal of Y.

Step 4: Verification

$$\begin{aligned} p_X(0) + p_X(1) &= \frac{3}{8} + \frac{5}{8} = 1 \\ p_Y(0) + p_Y(1) &= \frac{1}{2} + \frac{1}{2} = 1 \end{aligned}$$

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2 Correlation

2.1 Definition

$$R_{XY} = E[XY]$$

2.2 Interpretation

- Measures average value of product XY
- Computed about origin (not centered)

Important Observations

R_{XY} depends on:

- Mean values
- Spread of variables
- Dependence between variables

If $R_{XY} = 0 \Rightarrow$ Variables are orthogonal about origin

Limitation: This is not a clean measure of dependence because it mixes mean, spread, and dependence.

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3 Covariance

3.1 Definition

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Where:

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

3.2 Interpretation

Covariance measures how deviations from means vary together.

- $\mathrm{Cov}(X, Y) > 0 \to$ positive covariance (same direction)
- $\mathrm{Cov}(X, Y) < 0 \to$ negative covariance (opposite direction)
- $\mathrm{Cov}(X, Y) = 0 \to$ no linear co-variation

Important: Magnitude depends on units $\to$ cannot measure strength.

3.3 Algebraic Form (Detailed Derivation)

Start with definition:

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Step 1: Expand product

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Step 2: Apply expectation (linearity)

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Step 3: Substitute means

$$= E[XY] - \mu_Y\mu_X - \mu_X\mu_Y + \mu_X\mu_Y$$

Simplify:

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

3.4 Properties

1. Symmetry
   $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$
2. Variance
   $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$
3. Linearity
   $\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$
4. Independence
   If $X \perp Y \Rightarrow \mathrm{Cov}(X,Y) = 0$

Note: Zero covariance does not imply independence.

3.5 Example 1

Given: $Y = -2X + 3$
Find $\mathrm{Cov}(X,Y)$

Step-by-step Solution with Logic

Step 1: Use covariance formula

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Express XY in terms of X

$$XY = X(-2X + 3) = -2X^2 + 3X$$

Logic: Convert everything into X so expectations can be computed.
Step 3: Compute $E[XY]$

$$E[XY] = -2E[X^2] + 3E[X]$$

Step 4: Compute $E[Y]$

$$E[Y] = -2E[X] + 3$$

Step 5: Substitute

$$\mathrm{Cov}(X, Y) = (-2E[X^2] + 3E[X]) - E[X](-2E[X] + 3)$$

Step 6: Expand

$$= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X]$$

$$= -2E[X^2] + 2(E[X])^2$$

Step 7: Use variance identity

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

Thus:

$$\mathrm{Cov}(X, Y) = -2\,\mathrm{Var}(X)$$

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4 Pearson's Correlation Coefficient

4.1 Definition

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Where:

$$\sigma_X = \sqrt{\mathrm{Var}(X)}, \quad \sigma_Y = \sqrt{\mathrm{Var}(Y)}$$

4.2 Properties

- Measures direction and strength of linear relationship
- Normalized covariance
- Unit-free

Range:

$$-1 \leq \rho \leq 1$$

Interpretation:

- $\rho > 0 \to$ positive linear trend
- $\rho < 0 \to$ negative linear trend
- $\rho = 0 \to$ no linear correlation

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5 Example 2

Given:

$$f_{XY}(x, y) = \frac{xy}{96}, \quad 0 < x < 4,\ 1 < y < 5$$

Step 1: Compute $E[X]$

$$E[X] = \frac{1}{8}\int_0^4 x^2 dx = \frac{8}{3}$$

Step 2: Compute $E[Y]$

$$E[Y] = \frac{1}{12}\int_1^5 y^2 dy = \frac{31}{9}$$

Step 3: Compute $E[XY]$

$$E[XY] = E[X]E[Y] = \frac{248}{27}$$

Step 4: Compute $E[2X + 3Y]$

$$E[2X + 3Y] = \frac{47}{3}$$

Step 5: Compute Var(X)

$$\mathrm{Var}(X) = \frac{8}{9}$$

Step 6: Compute Var(Y)

$$\mathrm{Var}(Y) = \frac{92}{81}$$

Step 7: Compute Cov(X,Y)

$$\mathrm{Cov}(X, Y) = 0$$

Step 8: Compute Correlation

$$\rho = 0$$

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6 Jointly Gaussian Random Variables

6.1 Definition

A pair (X, Y) is jointly Gaussian if:

- Joint density has bivariate Gaussian form
- Marginals are Gaussian

6.2 Joint Gaussian PDF

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left(\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(y\right.\right.\right.$$

6.3 Interpretation

- Used in signal processing
- Used in communication systems
- Models noisy measurements

Example: CO_2 concentration and temperature
If $\rho > 0 \rightarrow$ higher CO_2 corresponds to higher temperature

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